4 Power Round 2023: A Perfectly Cromulent Power Question

Instructions: The power question is worth 50 points; each part's point value is given in brackets next to the part. To receive full credit, the presentation must be legible, orderly, clear, and concise. If a problem says "list" or "compute," you need not justify your answer. If a problem says "determine," "find," or "show," then you must show your work or explain your reasoning to receive full credit, although such explanations do not have to be lengthy. If a problem says "justify" or "prove," then you must prove your answer rigorously. Even if not proved, earlier numbered items may be used in solutions to later numbered items, but not vice versa. Pages submitted for credit should be NUMBERED IN CONSECUTIVE ORDER AT THE TOP OF EACH PAGE in what your team considers to be proper sequential order. PLEASE WRITE ON ONLY ONE SIDE OF THE ANSWER PAPERS. Put the TEAM NUMBER (not the team name) on the cover sheet used as the first page of the papers submitted. Do not identify the team in any other way.

In a sequence of n consecutive positive integers, where $n > 1$, an element of the sequence is said to be *cromulent* if it is relatively prime to all other numbers in the sequence. Every element of a sequence with $n = 2$ is cromulent because any two consecutive integers are relatively prime to each other.

1. List a sequence of three consecutive positive integers with

- **a.** exactly one cromulent element; [1 pt]
- **b.** three cromulent elements. [1 pt]

2. List a sequence of four consecutive positive integers with

- **a.** exactly one cromulent element; [1 pt]
- **b.** two cromulent elements. [1 pt]

3. List a sequence of five consecutive positive integers with

- **a.** exactly one cromulent element; [1 pt]
- **b.** exactly two cromulent elements; [1 pt]
- **c.** three cromulent elements. [1 pt]

4. Show that any two consecutive odd positive integers are relatively prime to each other. [2 pts]

5.
- **a.** Show that every sequence of three consecutive positive integers contains a cromulent element. [2 pts]
- **b.** Show that every sequence of four consecutive positive integers contains a cromulent element. [2 pts]
- **c.** Show that every sequence of five consecutive positive integers contains a cromulent element. [3 pts]

6. Find the maximum and minimum possible number of cromulent elements in a sequence of n consecutive positive integers with

- **a.** $n = 6$; [3 pts]
- **b.** $n = 7$. [3 pts]

7. Prove that there is at least one cromulent element in every sequence of n consecutive positive integers with

- **a.** $n = 8$; [4 pts]
- **b.** $n = 9$. [4 pts]

8.
- **a.** Find two sequences of 17 consecutive positive integers with no cromulent elements. [6 pts]
- **b.** Compute the least possible element of such a sequence of length 17. [2 pts]

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9. The goal of this problem is to prove the following claim:

Claim: For $k \geq 300$, there is a sequence of k consecutive integers with no cromulent elements.

Let $\pi(n)$ denote the number of primes less than or equal to n. You may use the following lemma without proof in the problems that follow.

Lemma: For $x \geq 75$, $\pi(2x) - \pi(x) \geq 2\lfloor \log_2(x) \rfloor + 2$.

One argument for proving the above claim begins as follows. Let $m = \lfloor \frac{k}{4} \rfloor \geq 75$, let $p_1, p_2, \ldots, p_r$ denote the primes in the set $\{1, 2, \ldots, m\}$ (note that $p_1 = 2$), and let p_{r+1}, p_{r+2}, $\ldots$, p_s denote the primes in the set $\{m+1, m+2, \ldots, 2m\}$.

a. Prove that if each of the k consecutive integers in the sequence is divisible by at least one of the primes $p_1, p_2, \ldots, p_s$, then the sequence has no cromulent elements. [2 pts]

Let x be a solution to the system of congruences $x \equiv 1 \pmod 2$ and $x \equiv 0 \pmod{p_2 p_3 \cdots p_r}$. Then the integers

$$x - 2m,\ x - 2m + 2,\ \ldots,\ x - 2,\ x,\ x + 2,\ \ldots,\ x + 2m - 2,\ x + 2m$$

form a sequence of $2m + 1$ consecutive odd integers of the form $x \pm 2y$, where y varies from 0 to m.

b. Prove that every number in the above sequence, except those in which y is a power of 2, is divisible by one of the primes $p_2, \ldots, p_r$. [2 pts]

Let $t = \lfloor \log_2(m) \rfloor + 1$. Then the elements of the above list not yet accounted for are $x \pm 2, x \pm 2^2, \ldots, x \pm 2^t$.

c. Prove that there are at least $2t$ primes in the set $\{m+1, m+2, \ldots, 2m\}$. [1 pt]

It therefore follows that $s - r \geq 2t$, hence $p_{r+2t} \leq p_s$.

d. Show that there exists an integer x such that

$$\begin{aligned} x &\equiv 1 \pmod 2, \\ x &\equiv 0 \pmod{p_2 p_3 \cdots p_r}, \end{aligned}$$

and for each $u \in \{1, \ldots, t\}$,

$$\begin{aligned} x + 2^u &\equiv 0 \pmod{p_{r+u}}, \text{ and} \\ x - 2^u &\equiv 0 \pmod{p_{r+t+u}}. \end{aligned}$$

[2 pts]

e. Let x be a solution to the system of congruences in part (d). Show that the sequence of $4m+3$ consecutive integers

$$x - 2m - 1,\ x - 2m,\ x - 2m + 1,\ \ldots,\ x - 1,\ x,\ x + 1,\ \ldots,\ x + 2m + 1$$

has no cromulent elements. [3 pts]

f. Show that for $k \geq 300$, there is a sequence of k consecutive integers with no cromulent elements. [2 pts]

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